HydroFlow HF Series Centrifugal Pumps Installation and Maintenance Manual

The HydroFlow HF series are single-stage, end-suction centrifugal pumps designed for clean water, cooling circuits and light industrial liquids. This manual covers the HF-100, HF-200 and HF-300 models. Read it completely before installing or operating the pump and keep it near the installation.

Every pump is hydrostatically tested at the factory at 1.5 times the maximum working pressure of 16 bar, that is at 24 bar, before it is shipped.

Intended use

The pumps may be used with liquids at temperatures between -10 °C and 90 °C. They must not be used for flammable or explosive liquids, or for liquids containing abrasive solids larger than 2 mm.

Warranty

The warranty period is 24 months from the date of commissioning or 30 months from the date of delivery, whichever comes first. The warranty is void if non-original spare parts are used or if the pump is operated outside the limits given in this manual.

Technical specifications

Table 1 lists the main ratings of each model at 50 Hz.

Model	Max flow (m3/h)	Max head (m)	Motor power (kW)	Weight (kg)
HF-100	12	32	2.2	38
HF-200	25	45	5.5	64
HF-300	40	58	11.0	102

The noise level does not exceed 72 dB(A) at a distance of 1 m for any model. All motors are IE3 premium efficiency, protection class IP55.

Installation

Mount the pump on a flat concrete foundation whose mass is at least 1.5 times the mass of the pump. Leave a free space of at least 0.5 m on all sides of the pump for service access.

The suction pipe must have a diameter at least equal to the pump inlet and should rise continuously towards the pump to avoid air pockets. Support all pipes independently so that no pipe forces act on the pump casing.

After installation check the coupling alignment. The shaft misalignment must not exceed 0.05 mm; larger values cause bearing wear and seal leakage.

Electrical connection

The motor must be protected by a thermal overload relay set to the rated motor current. Connect the earth conductor before any other wire. Check the direction of rotation by briefly starting the motor.

Performance curve

Figure 1 shows how the head of the HF-200 changes with the flow rate.

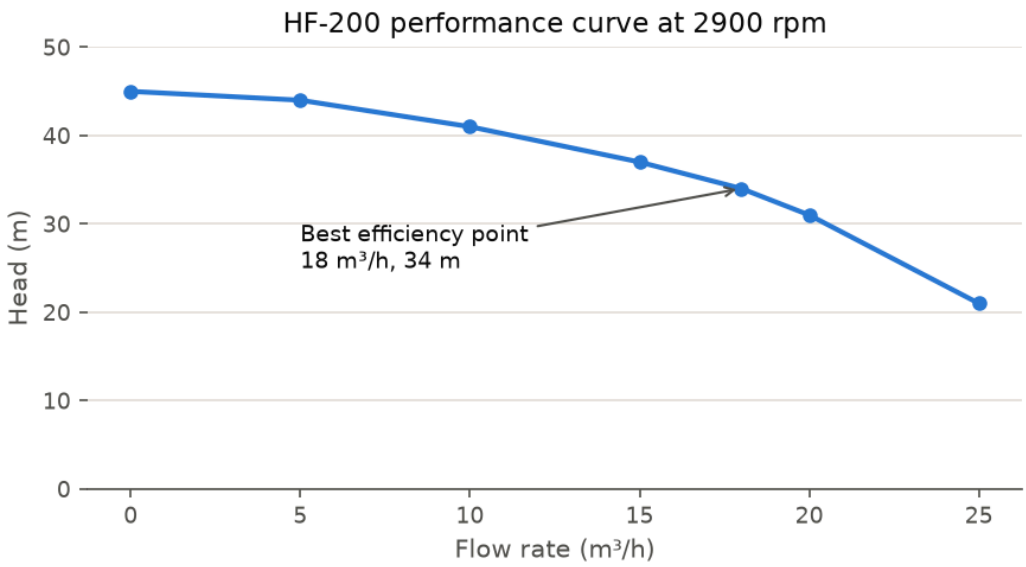


Figure 1: Performance curve of the HF-200 (head versus flow rate at 2900 rpm).

The best efficiency point (BEP) of the HF-200 is at 18 m3/h and 34 m head. For a long service life, operate the pump between 70% and 110% of the BEP flow.

Maintenance schedule

Task	Interval
Check mechanical seal for leaks	Weekly
Lubricate motor bearings	Every 2,000 operating hours
Inspect impeller and wear rings	Every 8,000 operating hours
Replace mechanical seal	Every 16,000 hours or when leaking

Use lithium-complex grease of NLGI grade 2 for the motor bearings, 15 g per bearing. Always isolate the motor from the power supply before any maintenance work.

TROUBLESHOOTING GUIDE

Problem: Pump does not deliver water.
Cause: Suction pipe and casing not primed.
Remedy: Fill the pump casing and suction pipe with liquid.

Problem: Excessive vibration.
Cause: Impeller out of balance or cavitation.
Remedy: Rebalance the impeller; increase the NPSH available.

Problem: Motor overheats.
Cause: Supply voltage deviates by more than 10 percent.
Remedy: Check the supply voltage.

Service hotline: +49 40 5550 1234